SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

***CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I LABHANA PRAVEEN/192426007 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 34 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-struct ured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization , incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

LABHANA PRAVEEN
192426007 3
CSA0710

21

22.10.26 COMPUTER NETWORKS QP 1 AT-1

1. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Introduction:

The transport Layer is the fourth layer of the OSI (open Systems Interconnection) model. It provides communication between applications running on different computers. It is responsible for data Segmentation, flow control, error recovery, and reliable data tranfer when a protocol such as TCP is used.

One important technique used by the Transport Layer is the sliding window protocol. It allows a sender to transmit multiple frames or data units before waiting for acknowledgements from the receiver.

Given Data:

⇒ Sliding window size = 6 frames

⇒ Total number of frames = 48

=> User data per frame = 1024 bytes.

=> Lost frames in each window = 1

We need to calculate:

1. Total number of transmission windows.
2. Total number of retransmissions.
3. The role of flow control in improving communi-cation efficiency.

1. Total number of transmission Windows.

Formula:

$$\text{Number of Windows} = \frac{\text{Total frames}}{\text{Window size}}$$

Substitution:

$$\text{Number of Windows} = \frac{48}{6}$$

= 8 windows

2. Total number of retransmissions:

Formula:

Retransmissions = Number of windows X Lost frames per window

Subsitition:

$= 8 \times 1$

$= 8$ retransmissions

3. Total transmission including retransmissions:

Formula:

Total transmissions = Orginal frame + Retransmissions

Subsitition:

$= 48 + 8$

$= 56$ transmissions

4. Total user data transmitted originally:

Formula:

Total data = Number of frames x Data per frame

Substition:

$= 47 \times 1024$

$= 49,152$ bytes

Convert into KB:

$= \frac{49,152}{1024}$

$= 48$ KB

2. A multimedia applications transfers a 600MB video file. The Session Layer inserts a synchronization checkpoint after every 60MB of transmitted data. During transmission, a failure occurs after 390MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Introduction:

The Session Layer is the fifth layer of OSI model. It is responsible for establishing, managing, synchronizing, and terminating communication sessions between applications on different devices. It helps maintain continous communication and supports recovery when interruptions occur.

Given Data:

⇒ Total video file Size = 600MB

⇒ Checkpoint interval = 60MB

⇒ Data transferred before failure = 390MB

We need to Calculate:

1. Number of checkpoints created before failure.
2. Amount of data that must be retransmitted after recovery.

3. Importance of synchronization in the Session Layer.

1. Calculation of the Number of checkpoints.

Formula:

$$\text{Number of checkpoints} = \frac{\text{Data transferred}}{\text{checkpoint interval}}$$

$$= \frac{390}{60}$$

$$= 6.5$$

- Checkpoint 1 = 60MB
- Checkpoint 2 = 120MB
- Checkpoint 3 = 180MB
- Checkpoint 4 = 240MB
- Checkpoint 5 = 300MB
- Checkpoint 6 = 360MB

Therefore,

Total checkpoints = 6

2. Calculation of the Data to Be Retransmitted.

Formula:

Data to retransmit = Data transfered before failure − Latest checkpoint

= 390 − 360

= 30MB

3. Importance of Synchronization in the Session Layer:

Supports Recovery After Failure:
Checkpoints allow the system to resume a transfer from a previously saved position instead of restarting the entire process.

Reduces Data Retransmission:
Without checkpoints, the applications might need to retransmit a much larger amount of data after a failure.

Saves Time:
Resuming from a saved checkpoint avoids repeating work that has already been completed. This is especially usefull when transferring large video files.

Improves Communications Efficiency.
Synchronization reduces unnecessary retransmissions and make recovery more efficient. This can save network bandwidth and reduce the time needed to complete the transfer.

3. A packet experiences the following processing delays at different OSI layers: Application Layer = 4ms, Transport process Layer = 3ms, Network Layer = 5ms, Data Link Layer = 2ms, and Physical Layer = 1ms. The propogation delay is 18ms, and the transmission delay is 12ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given Data:

A packet experiences processing delays at different OSI layers. The propagation delay is 18ms, and the transmissions delay is 12ms.

Application Layer = 4ms

Transport Layer = 3ms

Network Layer = 5ms

Data Link Layer = 2ms

Physical Layer = 1ms

Propagation Layer = 18ms

Transmission Delay = 12ms

Formula:

$$D_{total} = D_{processing} + D_{propagation} + D_{transmission}$$

To calculation of Processing Delay:

$$D_{processing} = 4 + 3 + 5 + 2 + 1$$

$$= 15ms$$

Calculation of Total End-to-End Delay:

$$D_{total} = 15 + 18 + 12$$

$$D_{total} = 45 ms$$

Therefore, the total end-to-end delay experienced by the packet is 45 milliseconds.

Explanation of Each OSI Layer:

1. Application Layer (4ms):

Provides network services to user applications, such as web browsing, email, and file transfer.

2. Transport Layer (3ms):

Divides Data into segments and provides end-to-end communication, error recovery, and reliable delivery when using TCP.

3. Network Layer (5ms):

Determines the appropriate route and forwards packets toward the destination using IP addressing

4. Data Link Layer (2ms):

Organizes packets into frames and performs functions such as MAC addressing and error detection.

5. Physical Layer (1ms):

Transmits bits through physical media such as cable, fiber optics, or wireless signals.

6. Propagation Layer (15ms):

The time taken by the signal to travel from the sender to the receiver through the transmission medium.

7. Transmission Delay (12ms):

The time required to place all the packet's bits onto the communication link.

4. A network transfers 80MB of useful data using frames of 1600 bytes. Where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead protocol overhead affects network performance.

Introduction:

A network transfer 80MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required.

Given Data:

=> Useful data = 80MB

=> Frame Size = 1600 bytes

=> Protocol overhead per frame = 80 bytes

Assume:

$1MB = 10^6$ bytes

$80MB = 80 \times 10^6$

Useful data = 80,000,000 bytes

To calculate the Useful Data per Frame:

Useful data per frame = Frame size - Overhead

= 1600 - 80

= 1520 bytes

To calculate the total Number of Frames:

$$\text{Number of frames} = \frac{\text{Total useful data}}{\text{Useful data per frame}}$$

$$= \frac{80,000,000}{1520}$$

= 52,631.58

Total frames: 52,632

To calculate the Total Protocol overhead:

Total overhead = Number of frames x Overhead per frame.

= 52,632 x 80

= 4,210,560 bytes.

Convert into MB:

$$= \frac{4,210,560}{10^6}$$

Total overhead ≈ 4.21 MB

To calculate Transmission Efficiency:

$$\text{Efficiency} = \frac{\text{Useful data}}{\text{Total transmitted data}} \times 100$$

Total transmitted data:

$$= 80,000,000 + 4,210,560$$

$$= 84,210,560 \text{ bytes}$$

Therefore,

$$\text{Efficiency} = \frac{80,000,000}{84,210,560} \times 100$$

Efficiency ≈ 95%

How protocol overhead Affects Network Performance

- Reduces efficiency
- Increases bandwidth usage
- May increase transmission time
- Improves reliability
- Supports communication.